

王业涵

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教育背景

华东师范大学

计算机科学与技术 | 本科 | GPA: 3.71/4.00

主修课程: 量子计算、数学分析、线性代数、离散数学、数据结构、算法分析与设计、数字逻辑电路、大学物理、概率论与数理统计、操作系统、机器学习等。

交流项目: 牛津大学暑期班 (师从 Prof. Kobi Kremnizer, 修读《Quantum Computing》, 成绩 97 分) 2025.08
青年使者赴美 (加州) 交流项目 2025.01-2025.02

实习经历

上汽大众 (SAIC VOLKSWAGEN) | 基于 RAG 的外购件质量专家模型

2026.07-2026.09

- 返库材料清洗: 面向多份 PPTX 与 PDF 格式的质量问题单, 分别用 python-pptx 与 PyMuPDF 按页抽取正文。再以预定义五字段为 Schema, 用 LLM 做约束式改写, 将原文归纳为车型、零件、问题概述、根本原因和遏制措施, 形成可按车型、零件筛选的案例表。
- 案例库 RAG 部署: 在结构化案例上构建 FAISS 与 BM25 混合检索, 经 RRF 融合与 Reranker 后生成回答, 同时开发上传预览入库、删除记录及索引热更新, 便于后续问题单进入知识库。
- 模块优化: 在“任务流程”模块补齐流程绑定、软删除功能和上手指引, 降低新人按规程流转的成本。

项目经历

基于 GitHub Pages 的个人主页 | 第一开发人

2026.08-2026.09

- 单页应用架构设计: 基于 Vite + React + TypeScript 构建单页应用, 采用长滚动首页与子路由相结合的结构, 内容与组件解耦, 以强类型 TypeScript schema 管理站点数据。
- 交互式地图与运动数据可视化: 集成 Leaflet 渲染 GPX 运动轨迹与 GeoJSON 旅行路线, 实现运动数据与生活足迹的动态地图展示, 支持缩放拖拽与轨迹动态绘制, 将原始 GPS 数据转化为可探索的交互式可视化体验。
- 隐私友好的数据分析与自动化部署: 接入 GoatCounter 实现无 Cookie 访问统计, 保护访客隐私; 配置 GitHub Actions 工作流, 实现代码推送后自动构建部署至 GitHub Pages, 支持持续集成与快速迭代。
- 项目已上线, 可通过 <https://visionnext100.github.io> 访问, 持续维护中。

医疗类 AI 产品 AdenoGuard——儿童腺样体智能筛查与辅助决策系统 | 第一开发人 2026.01-2026.03

- 真实临床数据与模型调优: 依托三甲医院授权的 10GB+ 真实多模态临床数据 (既往病史、腔镜影像、鼾声音频), 开发多模态 AI 引擎以协助筛查腺样体肿大病症。核心搭载 AutoGluon 训练框架, AUC 精度高达 0.91。
- 端云协同架构与小程序开发: 攻克跨模态数据流转难题, 实现多媒体医疗数据的无缝采集与全链路加密传输; 对接后端 FastAPI 接口, 保障云端 AI 推理快速响应; 使用 SQL 管理用户数据库。
- 医疗级闭环交互设计: 将诊断结果通过 CLAHE 图像增强技术及梅尔谱图可视化呈现, 完成从结构化病史录入到治疗辅助决策的业务闭环。利用 Canvas 渲染含置信度仪表盘、风险趋势图的医疗报告单, 赋能无创居家早筛。
- 目前已产出体验版小程序, 并申请软件著作权与发明专利。

竞赛经历

2026年 MCM/ICM 美国大学生数学建模竞赛 | 队长兼写手

2026.01-2026.02

- 数据清洗与特征工程: 针对分数归零、淘汰轮次缺失等问题设计标准化清洗策略, 通过分数归一化消除赛季差异, 并基于领域知识构造 “The Kingmaker Effect” 等高解释性特征。
- 混合建模与逆向优化: 综合运用 XGBoost、随机森林与 SLSQP 优化算法构建混合模型, 并结合 SHAP 进行特征归因; 构建逆向优化模型重构未知投票数据, 并通过敏感性分析验证模型稳健性。

OS华东区域赛 | 核心队员

2025.12-2026.01

- 内核级调度器设计: 针对传统 MLFQ 易被 CPU 密集型进程 “欺骗” 的漏洞, 基于 uCore OS 实现 Smart MLFQ 调度器; 引入 EWMA 实时评估 CPU 利用率, 动态识别 I/O 密集型与伪装型进程, 兼顾响应速度与抗欺骗能力。
- 系统压测与验证: 构造多类型进程压力测试, 对比 RR/Basic MLFQ/Smart MLFQ 的响应与周转时间, 验证 Smart MLFQ 在保持 I/O 低延迟的同时有效抑制欺骗行为。

荣誉奖项

- 2026年 美国大学生数学建模竞赛 (MCM/ICM) 二等奖 (Honorable Mention)
- 2026年 全国大学生计算机系统能力大赛-操作系统设计赛-华东区域赛-操作系统原理赛道 三等奖
- 2026年 第十六届全国大学生电子商务 “创新、创意及创业” 挑战赛全国总决赛 二等奖
- 2025年 华东师范大学优秀学生

专业技能

- 计算机能力: C++、Python、SQL、JavaScript、HTML、CSS、微信小程序开发
- 语言能力: 英语水平良好 (高考英语 143 分, CET-6: 568分)
- 办公软件: 熟练使用 Word、PowerPoint、Excel、飞书

Yehan Wang

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Education

East China Normal University

Shanghai, China

Bachelor of Computer Science and Technology | GPA: 3.71/4.00

Sep 2024 – Present

Relevant Courses: Quantum Computing, Mathematical Analysis, Linear Algebra, Discrete Mathematics, Data Structures, Introduction to Algorithm, Digital Logic Circuits, Physics, Probability and Statistics, Operating Systems, Machine Learning.

Exchange Programs:

·Oxford University Summer Institute – Quantum Computing (Advised by Prof. Kobi Kremizer) | Grade: 97/100 Aug 2025

·Youth Envoy Exchange Program – California, USA

Jan 2025 – Feb 2025

Internship

SAIC VOLKSWAGEN | A RAG-based expert model for the quality of purchased components

Jul 2026 – Sep 2026

- **Return-to-plant data cleaning:** For quality issue files in PPTX and PDF, extract page-level text with python-pptx and PyMuPDF respectively. Then run schema-constrained LLM rewriting against five predefined fields—vehicle, part, problem summary, root cause, and containment—to produce a case table filterable by vehicle and part.
- **Case-base RAG deployment:** Build hybrid FAISS and BM25 retrieval over the structured cases, fuse with RRF and a reranker, then generate answers. Also add upload-and-preview ingest, record deletion, and index hot-reload so later issue sheets can enter the knowledge base.
- **Module enhancement:** On the task-flow module, complete process binding, soft delete, and onboarding guidance to lower the cost for newcomers following standard procedures.

Projects

GitHub Pages personal site | Lead Developer

Aug 2026 – Sep 2026

- **Single-page architecture:** Built an SPA with Vite, React, and TypeScript, combining a long-scroll home page with nested routes. Decoupled content from components and managed site data with a strongly typed TypeScript schema.
- **Interactive maps and activity visualization:** Integrated Leaflet to render GPX workout tracks and GeoJSON travel routes, with pan, zoom, and animated path drawing, turning raw GPS logs into an explorable map experience.
- **Privacy-friendly analytics and automated deploy:** Added cookieless visit stats via GoatCounter. Configured GitHub Actions so pushes build and deploy to GitHub Pages, supporting CI and fast iteration.
- Live at <https://visionnext100.github.io> and under active maintenance.

AdenoGuard — a pediatric-adenoid AI screening and decision-support product | Lead Developer

Jan 2026 – Mar 2026

- **Clinical data and model tuning:** Used 10GB+ authorized multimodal data from a tertiary hospital—history, endoscopic images, and snore audio — to build a multimodal AI engine for adenoid hypertrophy screening. Trained with AutoGluon; AUC reached 0.91.
- **Device-cloud architecture and mini-program:** Solved cross-modal data flow so multimedia clinical inputs can be captured and sent end-to-end encrypted; connected a FastAPI backend for low-latency cloud inference; managed users with SQL.
- **Closed-loop clinical UX:** Presented findings with CLAHE-enhanced images and mel spectrograms, from structured history intake to decision support, and rendered Canvas reports with a confidence gauge and risk trend for at-home screening.
- Shipped an experience-edition mini-program; software copyright and invention patent applications filed.

Contests

MCM/ICM 2026 | Team Leader & Writer

Jan 2026 – Feb 2026

- **Data cleaning and feature engineering:** Designed a pipeline for zeroed scores and missing elimination rounds, normalized scores across seasons, and used domain knowledge to build high-interpretability features such as the Kingmaker Effect.
- **Hybrid modeling and inverse optimization:** Combined XGBoost, random forests, and SLSQP, with SHAP for attribution; reconstructed unknown votes by inverse optimization and checked robustness via sensitivity analysis.

CSCC 2025 Operating System Design Competition | Core Team Member

Dec 2025 – Jan 2026

- **Kernel scheduler design:** Implemented Smart MLFQ on uCore OS to stop CPU-bound jobs gaming classic MLFQ; used EWMA to separate I/O-bound from disguised CPU hogs while keeping response time low.
- **Stress tests and validation:** Built mixed-workload benchmarks and compared RR, basic MLFQ, and Smart MLFQ on response and turnaround time, showing Smart MLFQ keeps I/O latency low while limiting gaming.

Honors & Awards

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| • MCM/ICM Honorable Mention | 2026 |
| • Second Prize – National Finals of the 16th National Student Innovation, Creativity and Entrepreneurship Challenge | 2026 |
| • Third Prize – CSCC 2025 Operating System Design Competition (East China Region) | 2026 |
| • Outstanding Student Award – East China Normal University | 2025 |

Technical Skills

- Programming: C++, Python, Pytorch, JavaScript, WeChat Mini Program Development
- Languages: Proficient in English (Gaokao: 143/150, CET-6: 568)
- Office Software: Proficient in Word, PowerPoint, Excel, Lark